

The uniform-family cutoff theorem

Claude, with Daniel Murfet

2026-09-07

Abstract

Astra #9's R5 target. For every s -parametrised double series c with the COMMON envelope $|c_{ij}(s)|\rho^{i+j} \leq H(s) \leq C_0(1+s)^D e^{\beta s L}$ and every $N \geq 1$,

$$|Z_c(N) - S_c(N)| \leq \text{uniformConst} \cdot N^{-E} (1 + \log N), \quad E = \min\left(p_1 + \frac{M_1}{k_1}, p_2 + \frac{M_2}{k_2}\right),$$

where uniformConst depends only on $(\beta, b, \rho, C_0, L, D, h, k, M)$: the constant is chosen before the coefficient family (`twoD_cutoff_uniform`). It is the explicit cutoff constant of unit 131 with every face constant replaced by its envelope bound (unit 132) at the face envelopes of the analytic adapter, plus the explicit mixed constant of unit 130. Zero `sorry/axiom`.

1 The things formalised

```
noncomputable def uniformConst ( b C L : ) (h h k k M M D : ) : :=
  ( i Finset.range M,
    faceEnvConst _i b h k k M
      (C * ( ^ i ) ^ 1 * (1 + ^ 1) ^ M) (C * ( ^ i ) ^ 1 * ( ^ M ) ^ 1 / (1 - b / ))
      * envMoment ( _i + (M - _i) / k ) 1 (gaussEnv L D))
+ ( j Finset.range M, ... symmetric ...)
+ ( i Finset.range M, j Finset.range M,
    faceEnvConst _j b (h + i) k k (M + k * M) (C * ( ^ (i + j) ) ^ 1) 0
      * envMoment ( _j + (M + k * M - _ij) / k ) 1 (gaussEnv L D))
+ C * ( ^ (M + M) ) ^ 1 / (1 - b / ) ^ 2 * Real.exp ((D : ) ^ 2 / )
  * mixConst ( / 2) (2 * L) b (h + M) (h + M) k k
theorem twoD_cutoff_uniform ( b p p C L : ) (h h k k M M D : )
  (h : 0 < ) (hb : 0 < b) (hb : b < ) (hk : 0 < k) (hk : 0 < k)
  (hp : ((h : ) + 1) / k = p) (hp : ((h : ) + 1) / k = p)
  (hM : p + ((M : ) - 1) / k < p + (M : ) / k)
  (hM : p + ((M : ) - 1) / k < p + (M : ) / k)
  (c : x → → ) (hcc : ij, Continuous (c ij)) (H : → ) (hH : Continuous H)
  (hc : ij s, |c ij s| * ^ (ij.1 + ij.2) H s) (hC : 0 C)
  (henv : s, 0 s → H s C * (1 + s) ^ D * Real.exp ( * s * L)) (N : )
  (hN : 1 N) :
|twoDamp b N h h k k (anaAmp c b)
  - mergedSum b h h k k M M (anaFaceU c b) (anaFaceV c b)
  (fun i j s => c (i, j) s) N|
uniformConst b C L h h k k M M D
  * N ^ (-(min (p + (M : ) / k) (p + (M : ) / k))) * (1 + Real.log N)
```

2 Mathematics

The analytic adapter supplies, for the u -face i , the Taylor-data envelope $|c_{im}(s)| \leq C_0 \rho^{-i} (1 + \rho^{-1})^{M_2} (1 + s)^D e^{\beta s L}$ ($m < M_2$) and the remainder envelope $H_i(s) = H(s) \rho^{-i-M_2} / (1 - b/\rho) \leq C_0 \rho^{-i-M_2} (1 - b/\rho)^{-1} (1 + s)^D e^{\beta s L}$; for the corners the constant Taylor datum c_{ij} has envelope $C_0 \rho^{-i-j} (1 + s)^D e^{\beta s L}$ and zero remainder. Unit 132 converts each face constant into `faceEnvConst` times one Gaussian envelope moment at the face's remainder exponent; the mixed constant is already envelope-only. Summing termwise gives `cutoffConst` $\leq$ `uniformConst`, and unit 131 gives the bound for every $N \geq 1$.

3 Paper-facing meaning

This is the uniformity Astra #9 asked for: a single remainder constant for the whole family of chart amplitudes $\eta_w e^{\beta s \xi_w}$ whose coefficient arrays share a weighted bound, i.e. for ξ, η analytic on a common neighbourhood of the closed box times the support of the partition of unity in the noncritical coordinates w . With $K(w) \equiv$ `uniformConst` measurable and bounded, the per-stratum integral over w can be fed to `param_integration` (unit 99) once the leading-order corollary is extracted and the coefficient functions $A(w), B(w)$ are shown measurable; that is the noncritical-coordinate adapter of Astra #6 rank 4.

4 Lean notes

- The hypothesis-discharge block of the analytic adapter (envelopes, remainders, mixed bound, `FaceMoments`) was reused verbatim by extracting it programmatically from the source file; the new theorem differs only in what is done with those hypotheses afterwards.
- Passing `_` for the envelope constants `Cf`, `CH` in `faceConst_le_env` lets unification read them off the target `uniformConst` summand; only the two envelope side goals then need proofs.
- `anaCoeff_env` produces the bound in the form `C * (1 + s)^i * (1 + s)^M * (1 + s)^D * exp`; `this.trans (le_of_eq (by unfold gaussEnv; ring))` reassociates it to `Cf * gaussEnv`.
- The whole unit compiled on the first attempt.